Topic 2: Interaction Design Process and Practice

SECV2113 Human-Computer Interaction

Faculty of Computing
Universiti Teknologi Malaysia

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02 SOME PRACTICAL ISSUES

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WHAT IS INVOLVED IN INTERACTION DESIGN

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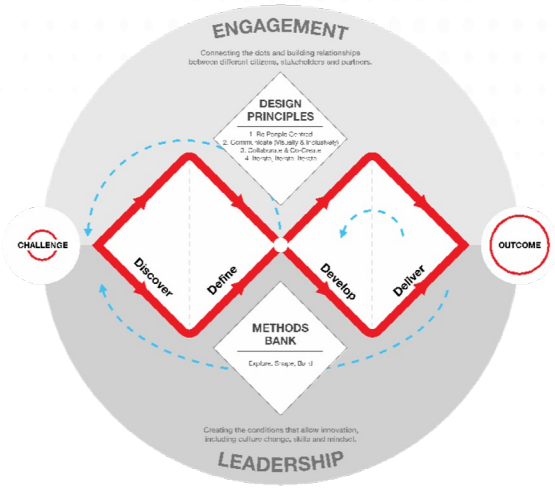
What is Involved in ID?

- It is a process:
  - Focused on discovering requirements, designing to fulfil requirements, producing prototypes and evaluating them
  - Focused on users and their goals
  - Involves trade-offs to balance conflicting requirements
- Generating alternatives and choosing between them is key
- Four approaches: user-centered design, activity-centered design, systems design, and genius design

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The Double Diamond of Design

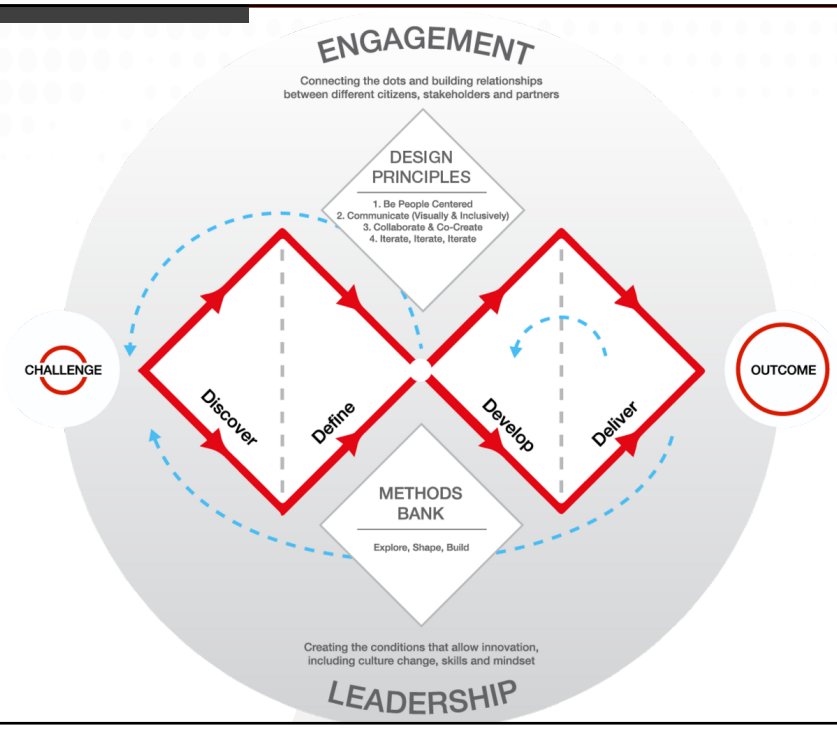


Focus:
General design thinking

Source: Adapted from www.designcouncil.org.uk/news-opinion/what-framework-innovation-design-councils-evolved-double-diamond

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The process: using the Double Diamond

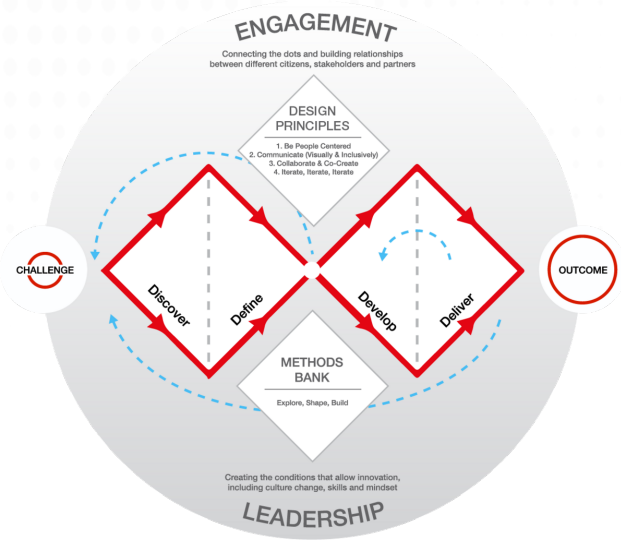
Design Council's Double Diamond clearly conveys a design process to designers and non-designers alike. The two diamonds represent a process of exploring an issue more widely or deeply (divergent thinking) and then taking focused action (convergent thinking).

•**Discover.** The first diamond helps people understand, rather than simply assume, what the problem is. It involves speaking to and spending time with people who are affected by the issues.

•**Define.** The insight gathered from the discovery phase can help you to define the challenge in a different way.

•**Develop.** The second diamond encourages people to give different answers to the clearly defined problem, seeking inspiration from elsewhere and co-designing with a range of different people.

•**Deliver.** Delivery involves testing out different solutions at small-scale, rejecting those that will not work and improving the ones that will.



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Who to Involve in ID?

- Identify stakeholders for the product
  - Individuals or groups who influence or are influenced by the success or failure of the project
  - Users are not the only stakeholders
  - Direct and indirect users, developers, executives, legislators
- Involving stakeholders is not always easy
  - May not need to involve all groups
  - Knowing the stakeholder groups is only the first step!

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Degrees of User Involvement

- Member of the design team
- Small group or individual activities
- Online contributions from thousands of users
  - Online Feedback Exchange (OFE) systems
  - Crowdsourcing design ideas
  - Citizen engagement
- Participatory design
- User involvement after product release
  - A/B testing
  - Customer reviews

What is People-centered Approach?

- **Three principles** for "useful & easy-to-use" products
  - Early focus on users and tasks
  - Empirical measurement
  - Iterative design
- Expanded to include
  1. People's tasks and goals drive development
  2. People's behaviour and context is studied
  3. People's characteristics are designed for
  4. Stakeholders are consulted throughout
  5. Design decisions are taken in this context
  - Specific usability and user experience goals
  - Iteration through all interaction design activities

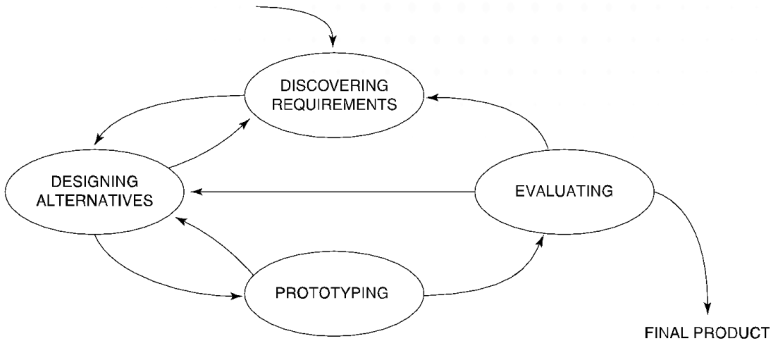
Four Basic Activities of ID

1. Discovering requirements
2. Designing alternatives
3. Prototyping alternative designs
4. Evaluating product and its user experience throughout

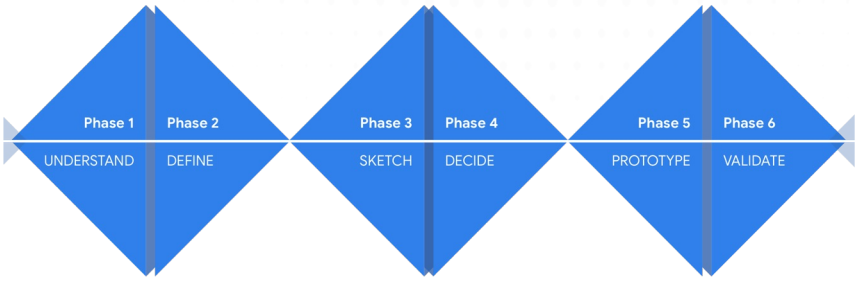
A Simple ID Lifecycle Model

Exemplifies a people-centered design approach

Focus:
Interaction design & usability



Another Lifecycle Model: Design Sprints

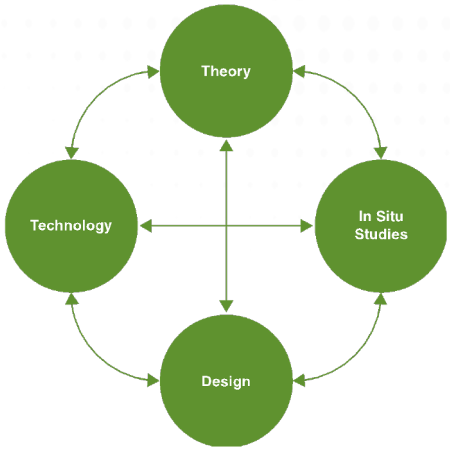


Source: designsprintkit.withgoogle.com/methodology/overview

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Another Lifecycle Model: Research in the Wild



A framework for research in the wild studies

Source: Rogers and Marshall, 2017, p6. (used courtesy of Morgan and Claypool)

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Comparison of HCI Lifecycle Models

Aspect	Simple ID Lifecycle	Double Diamond	Design Sprint	Research in the Wild
Main Goal	Usable interactive systems	Find problem & solution	Rapid prototype & test	Real-world understanding
Origin	HCI	Design Council (UK)	Google Ventures	HCI Research
Structure	Req → Design → Proto → Eval	Discover → Define → Develop → Deliver	Understand → Sketch → Decide → Proto → Test	Design → Deploy → Observe → Analyze
Duration	Flexible	Flexible	5 days	Long-term
Iteration	Yes	Yes	After sprint	Continuous
Focus	Usability	Creativity & framing	Fast validation	Real context & behavior
Environment	Controlled/lab	Flexible	Test sessions	Real world
Users	Participants	Stakeholders	5-7 users	Real users
Data	Usability metrics	Insights	Feedback	Contextual data
Output	Improved interface	Defined direction	Tested prototype	Research insights
Strength	Structured, usability	Broad thinking	Fast & collaborative	High realism
Limitation	Slow	Abstract	Shallow depth	Hard to control
Best For	HCI systems	Complex design	Early testing	Real-world use

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SOME PRACTICAL ISSUES

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Some Practical Issues

- How to find out what people need
- How to decide what to design
- How to generate alternative designs
- How to choose among alternatives
- How to integrate interaction design activities with other lifecycle models

How to find out what people need

- Users rarely know what is possible
- Instead:
  - Explore the problem space
  - Investigate potential users and their activities
  - Try out ideas with potential users
- Focus on peoples' goals, usability, and user experience goals, rather than expect stakeholders to articulate requirements

How to decide what to design

- Explore
  - What is the current user experience?
  - Why is a change needed?
  - How will this change improve the situation?

- Articulating the problem space
  - Team effort
  - Explore different perspectives
  - Avoid incorrect assumptions and unsupported claims

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How to generate alternative designs

- Humans tend to stick with something that works
- Considering alternatives helps identify better designs
- Where do alternative designs come from?
  - 'Flair and creativity': research and synthesis
  - Cross-fertilisation of ideas from different perspectives
  - Users can generate different designs
  - Prompts to provoke thinking, e.g. SCAMPER
  - Seek inspiration: other products and domain
  - Different perspectives
- Balancing constraints and trade-offs

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How to choose among alternatives

- Interaction design focuses on externally-visible and measurable behavior
- Technical feasibility
- Evaluation with users or peers
  - Prototypes not static documentation because behavior is key
- A/B Testing
  - Online method to inform choice between alternatives
  - Nontrivial to set appropriate metrics and choose user group sets
- Quality thresholds
  - Different stakeholder groups have different quality thresholds
  - Usability and user experience goals lead to relevant criteria

How to integrate ID activities within other models

- Integrating interaction design activities in lifecycle models from other disciplines requires careful planning
- Software development lifecycle models are prominent
- Integrating with agile software development:
  - It incorporates tight iterations
  - It champions early and regular feedback
  - It handles emergent requirements
  - It aims to strike a balance between flexibility and structure

AGILEUX

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Technical Debt in UX

- Making compromises that are expedient in short term, but cause problems in long term
- Refactoring to correct pragmatic trade-offs
- UX debt arises from trade-offs
  - A history of neglecting UX
  - Large portfolio of products developed independently

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Agile Development

- Short (one to three week) timeboxes of iterative development (sprint, iteration, cycle)
- Early and repeated customer/user feedback
- Re-prioritization of work based on customer or user feedback so that emergent requirements can be handled
- Many approaches, for example, eXtreme Programming (XP), Scrum, Kanban, and DSDM

AgileUX

- Integrates techniques from interaction design and agile methods
- AgileUX requires balancing research and reflection for good UX with rapid iterations incorporating user feedback
- In Agile iterations, requirements are elaborated and re-prioritised, rather than specified up-front

AgileUX (cont.)

- All techniques in UX are still relevant, but when to use and how much needs careful planning
- Focus on product, not design, as deliverable
- Cross-functional teams
- Three practical areas: user research, aligning work practices, and documentation

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User Research

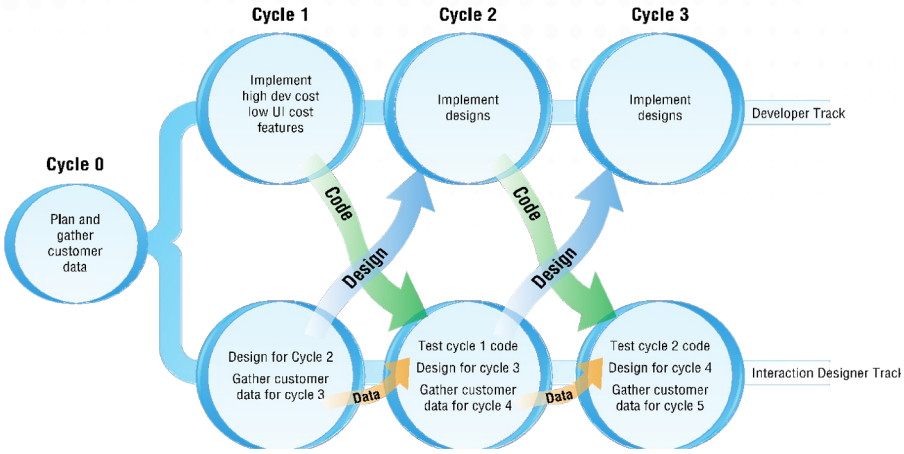
- Seeks to characterise users, tasks, and context through data gathering and analysis
- Detailed user research cannot be fitted within a limited timebox
  - Some user research can be performed in iteration 0 (zero), before implementation starts
  - Ongoing program of user research over a longer period of time

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Aligning Work Practices

- Designing a complete product upfront causes problems because of re-prioritization
- Some upfront work is needed (technical and UX)
- Use a dual tracks approach:
  - Create product vision before development starts
  - Do design work one iteration ahead of development
  - Some teams work two iterations ahead

Dual Tracks Approach to AgileUX

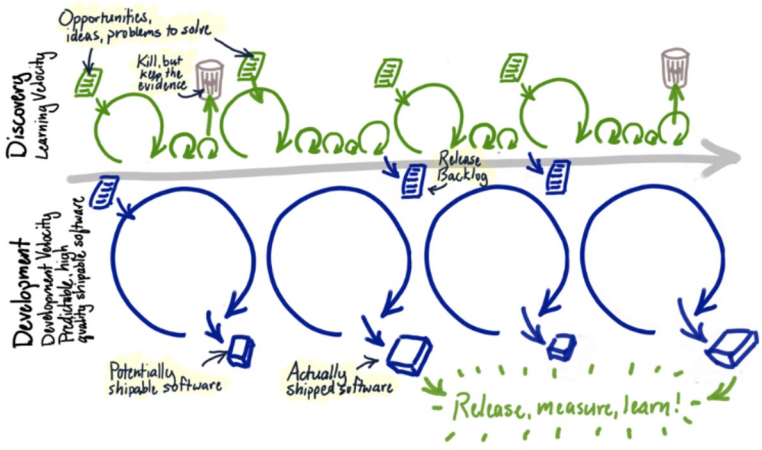


Source: Sy, D. (2007) Adapting usability investigations for development, Journal of Usability Studies 2(3), May, 112–130.

Aligning Work Practices (cont.)

- Advantages of dual tracks approach:
  - No design time wasted on features not implemented
  - Usability testing and contextual inquiry could be done on the same customer visit, saving time
  - Timely feedback on the designs was received from developers and customers
- Agile flexibility supports schedule changes if a problem is found
- Parallel tracks is commonly used

Dual Tracks: Discovery & Development



Source: www.jpattonassociates.com/dual-track-development

Documentation

- Common communication approach for UX designers
- Agile encourages minimal documentation so more time can be spent on design and discussion
- Only use documentation where needed. Ask:
  - Can the amount of time spent on documentation be reduced?
  - Who uses documentation?
  - What is the minimum needed by readers?
  - How efficient is the sign-off process?
  - Is there duplication anywhere?
  - How polished does documentation need to be?

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DESIGN PATTERNS

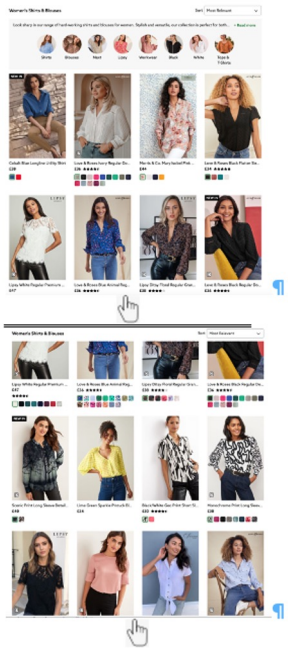
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Design Patterns

- Capture design experience:
  - A solution to a problem in a context
  - Can be instantiated in many ways: generative
- Patterns may be individual, in languages, in catalogues, galleries, or libraries
- Patterns often are associated with software components, for example, GitHub or platform websites

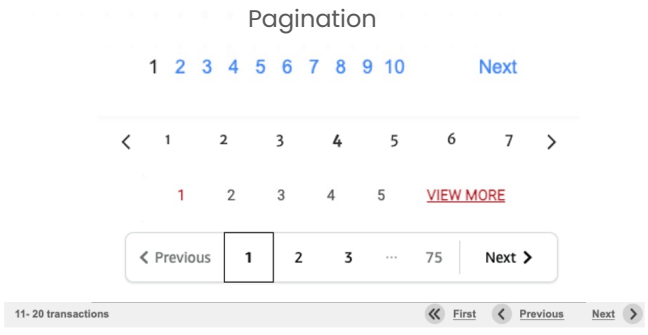
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Design Patterns



Continuous scrolling

Pagination



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Bad Design Patterns

- Capture design experience — but that doesn't necessarily mean good design:
  - Anti-patterns: Don't do it this way! (classing e.g.: "click here")
  - Dark patterns: deliberate tricks

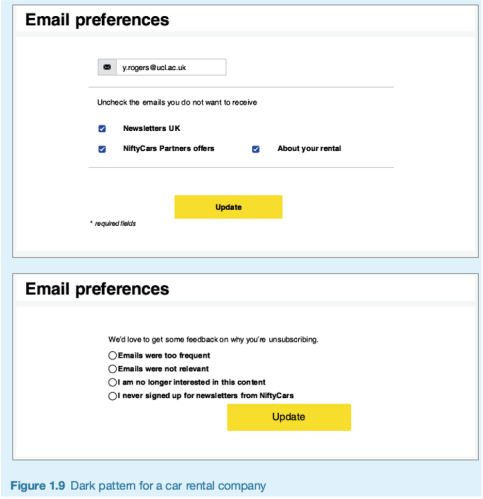


Figure 1.9 Dark pattern for a car rental company

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Design Language and Systems

- Structured collection of patterns and components
- May also include brand or accessibility guidelines
- Reduce effort, support learning between designers and increase collaboration
- Larger chunks of design can be reused
- User flows, e.g. overflow.io or uxarchive.com)

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OPEN SOURCE RESOURCES & TOOLS FOR ID

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Open Source Resources

- Components, frameworks, systems available free of charge
- Community-driven
- Available for interaction design:
  - Design pattern libraries
  - Bootstrap framework

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Tools for Interaction Design

- Tools support all aspects of the design process:
  - Creativity, sketching, simulation, brainstorming, library search, mind mapping, and video capture
- Tools integrate together to speed up prototyping
- Tooling landscape changes all the time
- Interactive wireframes or mockups can be produced using, for example:
  - Sketch©
  - Balsamiq©
  - Axure©
  - Figma©
- Higher-fidelity prototypes can be produced by linking interactive wireframes to design pattern library with software components

Summary (1)

- Four basic activities in interaction design process
  - Discovering requirements
  - Designing alternatives
  - Prototyping
  - Evaluating
- People-centered design rests on three principles
  - Early focus on users and tasks
  - Empirical measurement
  - Iterative design
- Understand problem space before starting design
- Involve stakeholder groups throughout development
- Generate alternative designs and evaluate them

Summary (2)

- AgileUX refers to approaches that integrate UX design and agile development
  - Needs a balance between research and reflection, and rapid iterations
  - Requirements are repeatedly re-prioritized, which seeks to avoid wasted effort
  - UX design activities need careful planning: when, how much, and how to take forward
- Design patterns and implementation libraries support good user experience design (look out for anti- and dark patterns)
- Open source resources, for example, on GitHub, make development of standard applications easier and quicker
- A range of digital tools to support interaction design in practice is available